

Study Guide and Review



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Be sure the following Key Concepts are noted in your Foldable.

	Words	Example(s)
A Plan for Problem Solving		
+ & - of Integers		
x & ÷ of Integers		
Solving + & - Equations		
Solving x & ÷ Equations		

Key Concepts

Order of Operations (Lesson 1-2)

1. Do all operations within grouping symbols first.
2. Evaluate all powers before other operations.
3. Multiply and divide in order from left to right.
4. Add and subtract in order from left to right.

Operations With Integers (Lessons 1-4 to 1-6)

- To add integers with the same sign, add their absolute values. The sum has the same sign as the integers.
- To add integers with different signs, subtract their absolute values. The sum has the sign of the integer with the greater absolute value.
- To subtract an integer, add its opposite or additive inverse.
- The product or quotient of two integers with the same sign is positive.
- The product or quotient of two integers with different signs is negative.

Solving Equations (Lessons 1-9, 1-10)

- If you add or subtract the same number to/from each side of an equation, the two sides remain equal.
- If you multiply or divide each side of an equation by the same nonzero number, the two sides remain equal.

Key Vocabulary

absolute value (p. 36)	integer (p. 35)
additive inverse (p. 43)	inverse operations (p. 66)
algebra (p. 29)	negative number (p. 35)
algebraic expression (p. 29)	numerical expression (p. 29)
coordinate (p. 35)	opposites (p. 43)
counterexample (p. 31)	order of operations (p. 29)
define a variable (p. 57)	powers (p. 30)
equation (p. 57)	solution (p. 65)
evaluate (p. 29)	solve (p. 65)
inequality (p. 35)	variable (p. 29)

Vocabulary Check

State whether each sentence is *true* or *false*. If *false*, replace the underlined word or number to make a true sentence.

1. Operations that “undo” each other are called order of operations.
2. The symbol for absolute value is $| \quad |$.
3. A mathematical sentence that contains an equals sign is an inequality.
4. An integer is a number less than zero.
5. A property is an example that shows that a conjecture is false.
6. The value of the variable that makes the equation true is called the solution.
7. The number that corresponds to a point is called its coordinate.
8. A power is a symbol, usually a letter, used to represent the number.
9. An expression that contains a variable is an algebraic expression.

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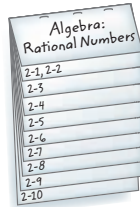


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Key Concepts

Rational Numbers (Lesson 2-1)

- A rational number is any number that can be expressed in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

Multiplying and Dividing Fractions

(Lessons 2-3 and 2-4)

- To multiply fractions, multiply the numerators and multiply the denominators.
- To divide by a fraction, multiply by its multiplicative inverse.

Adding and Subtracting Fractions

(Lessons 2-5 and 2-6)

- To add or subtract fractions, rename the fractions using the least common denominator. Then add or subtract and simplify, if necessary.

Powers and Scientific Notation

(Lessons 2-9 and 2-10)

- A number is expressed in scientific notation when it is written as the product of a factor and a power of 10. The factor must be greater than or equal to 1 and less than 10.

Key Vocabulary

bar notation (p. 85)	power (p. 126)
base (p. 126)	rational number (p. 84)
dimensional analysis (p. 98)	reciprocals (p. 102)
exponent (p. 126)	repeating decimal (p. 85)
like fraction (p. 108)	scientific notation (p. 130)
multiplicative inverses (p. 102)	terminating decimal (p. 85)
	unlike fraction (p. 114)

Vocabulary Check

State whether each sentence is *true* or *false*. If *false*, replace the underlined word or number to make a true sentence.

- Like fractions have the same denominator.
- The number that is expressed using an exponent is a rational number.
- Dimensional analysis is the process of including units of measurement in computation.
- The number $0.\overline{3}$ is a repeating decimal.
- Numbers that can be written as fractions are called reciprocals.
- The number 4.05×10^8 is written in bar notation.
- The number 2.75 is a terminating decimal.
- The base tells how many times a number is used as a factor.
- Two numbers whose product is 1 are multiplicative inverses of each other.
- The number 5^4 is a power.

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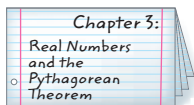


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Key Concepts

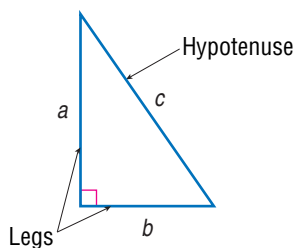
Square Roots and Irrational Numbers

(Lessons 3-1, 3-2, and 3-4)

- A square root of a number is one of its two equal factors.
- An irrational number is a number that cannot be expressed as $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

Pythagorean Theorem (Lessons 3-5 to 3-7)

- In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the legs.



- If the sides of a triangle have lengths a , b , and c units such that $c^2 = a^2 + b^2$, then the triangle is a right triangle.

Key Vocabulary

coordinate plane (p. 173)
hypotenuse (p. 162)
irrational number (p. 155)
legs (p. 162)
perfect square (p. 144)
Pythagorean Theorem (p. 162)
radical sign (p. 144)
real number (p. 155)
square root (p. 144)
Venn diagram (p. 152)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or
number to make a true sentence.

1. The number 11 is a perfect square.
2. The symbol that is used to indicate a square root is the radical sign.
3. A real number is a number that cannot be expressed as the quotient of two integers.
4. If the measures of the sides of a triangle are 6 inches, 8 inches and 10 inches, then the triangle is a right triangle.
5. The opposite of squaring a number is finding a converse.
6. A Venn diagram uses overlapping rectangles to organize information and solve problems.
7. The hypotenuse is the shortest side of a right triangle.
8. The Pythagorean Theorem states that the sum of the squares of the lengths of the legs of a right triangle equals the square of the length of the hypotenuse.

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Key Concepts

Proportions (Lessons 4-2 and 4-3)

- If two related quantities are proportional, then they have a constant ratio.
- A proportion is an equation stating that two ratios or rates are equivalent.
- The cross products of a proportion are equal.

Similar Polygons (Lesson 4-5)

- If two figures are similar with a scale factor of $\frac{a}{b}$, then the perimeters of the figures have a ratio of $\frac{a}{b}$.

Measurement (Lessons 4-6 and 4-8)

- You can multiply by a unit ratio to *convert* or change from larger units to smaller units.
- To convert from smaller units to larger units, multiply by the *reciprocal* of the appropriate unit ratio.
- A scale is determined by the ratio of a given length on a drawing or model to its corresponding actual length.

Rates (Lessons 4-1, 4-9, and 4-10)

- A rate is a comparison of two quantities with different types of units.
- To find the rate of change, divide the difference in the y -coordinates by the difference in the x -coordinates.
- Two quantities a and b have a proportional linear relationship if they have a constant ratio and a constant rate of change.

Key Vocabulary

congruent (p. 207)	proportional (p. 194)
constant of proportionality (p. 200)	rate (p. 191)
constant rate of change (p. 236)	rate of change (p. 230)
corresponding parts (p. 206)	ratio (p. 190)
cross products (p. 198)	scale (p. 224)
equivalent ratios (p. 198)	scale drawing (p. 224)
linear relationship (p. 236)	scale factor (p. 207)
nonproportional (p. 194)	scale model (p. 224)
polygon (p. 206)	similar (p. 206)
proportion (p. 198)	unit rate (p. 191)
	unit ratio (p. 214)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or number to make a true sentence.

1. Polygons that have the same size are called similar polygons.
2. A unit ratio is one in which the denominator is 1 unit.
3. A ratio of two measurements having similar units is called a rate.
4. In a relationship in which the ratio is not constant, the two quantities are said to be nonproportional.
5. A scale is the ratio of a length on a drawing or model to its corresponding actual length.
6. Comparing two numbers by multiplication is called a proportion.

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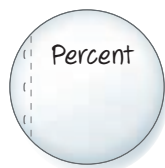


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Key Concepts

Percent (Lessons 5-1 and 5-2)

- A percent is a ratio that compares a number to 100.
- To write a percent as a decimal, divide by 100 and remove the percent symbol.
- To write a decimal as a percent, multiply by 100 and add the percent symbol.

Percent Proportion (Lesson 5-3)

- A percent proportion is $\frac{\text{part}}{\text{whole}} = \text{percent}$, where the percent is written as a fraction.

Percent-Fraction Equivalents			
$25\% = \frac{1}{4}$	$50\% = \frac{1}{2}$	$75\% = \frac{3}{4}$	$100\% = 1$
$20\% = \frac{1}{5}$	$40\% = \frac{2}{5}$	$60\% = \frac{3}{5}$	$80\% = \frac{4}{5}$
$16\frac{2}{3}\% = \frac{1}{6}$	$33\frac{1}{3}\% = \frac{1}{3}$	$66\frac{2}{3}\% = \frac{2}{3}$	$83\frac{1}{3}\% = \frac{5}{6}$
$12\frac{1}{2}\% = \frac{1}{8}$	$37\frac{1}{2}\% = \frac{3}{8}$	$62\frac{1}{2}\% = \frac{5}{8}$	$87\frac{1}{2}\% = \frac{7}{8}$
$10\% = \frac{1}{10}$	$30\% = \frac{3}{10}$	$70\% = \frac{7}{10}$	$90\% = \frac{9}{10}$

Percent Equation (Lesson 5-7)

- A percent equation is $\text{part} = \text{percent} \cdot \text{whole}$, where the percent is written as a decimal.

Percent of Change (Lesson 5-8)

- A percent of change is a ratio that compares the change in quantity to the original amount.

Key Vocabulary

compatible numbers (p. 275)	percent of change (p. 284)
compound interest (p. 294)	percent of decrease (p. 285)
discount (p. 286)	percent equation (p. 279)
interest (p. 290)	percent of increase (p. 285)
markup (p. 286)	percent proportion (p. 263)
percent (p. 252)	principal (p. 290)
	selling price (p. 286)

Vocabulary Check

Choose the correct term or numbers to complete each sentence.

1. A (proportion, percent) is a ratio that compares a number to 100.
2. (Percents, Compatible numbers) are numbers that are easy to divide mentally.
3. A (markup, discount) is an increase in price.
4. 25% of 16 is (4, 40).
5. The (interest, principal) is the amount borrowed.
6. In the proportion $\frac{6}{5} = \frac{120}{100}$, the (part, whole) is 6.
7. A (markup, discount) is a decrease in price.
8. The interest formula is ($I = prt$, $p = Irt$).
9. The number 0.015 written as a percent is (0.15%, 1.5%).
10. The (interest, principal) is the money paid for the use of money.

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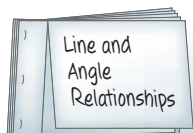


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Key Concepts

Angle Relationships (Lesson 6-1)

- The sum of the measures of complementary angles is 90° .
- The sum of the measures of supplementary angles is 180° .
- Vertical angles are opposite angles formed by intersecting lines and are congruent.

Polygons (Lessons 6-3 and 6-4)

- The sum of the measures of the interior angles of a polygon is $(n - 2)(180)$.
- In congruent polygons, corresponding sides and angles are congruent.

Symmetry (Lesson 6-5)

- A figure with line symmetry can be folded over a line so that the two halves match.
- A figure with rotational symmetry can be rotated about its center so that it looks exactly as it does in its original position.

Transformations (Lessons 6-6 and 6-7)

- In a reflection, the image is congruent to the original figure, but the orientation of the image is different from that of the original figure.
- In a translation, the image is congruent to the original figure, and the orientation of the image is the same as that of the original figure.

Key Vocabulary

angle (p. 307)	line symmetry (p. 327)
collinear (p. 306)	plane (p. 306)
complementary angles (p. 307)	point (p. 306)
congruent angles (p. 306)	ray (p. 307)
image (p. 332)	reflection (p. 332)
inductive reasoning (p. 314)	rotational symmetry (p. 328)
interior angles (p. 316)	supplementary angles (p. 307)
line (p. 306)	transformation (p. 332)
line of reflection (p. 332)	translation (p. 337)
line of symmetry (p. 327)	vertical angles (p. 307)

Vocabulary Check

State whether each sentence is *true* or *false*.

If *false*, replace the underlined word or number to make a true sentence.

1. $m\angle 1$ is read as the measure of $\angle 1$.
2. A translation of $(3, -2)$ means a translation 3 units left and 2 units down.
3. If $\triangle ABC \cong \triangle DEF$, then $\angle C \cong \angle E$.
4. Inductive reasoning is the process of making a rule after observing several examples and using that rule to make a decision.
5. A polygon whose angles are all congruent is said to be equilateral.
6. A rectangle will match itself after being rotated 90° , 180° , and 270° .
7. P' , the image of point P , is read as P prime.
8. When a transformation occurs, the resulting figure is called a line of reflection.

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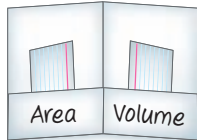


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Key Concepts

Circles (Lesson 7-1)

- Circumference: $C = \pi d$ or $C = 2\pi r$
- Area: $A = \pi r^2$

Volume (Lessons 7-5 and 7-6)

- Prism: $V = Bh$
- Cylinder: $V = Bh$ or $V = \pi r^2 h$
- Pyramid: $V = \frac{1}{3}Bh$
- Cone: $V = \frac{1}{3}Bh$ or $V = \frac{1}{3}\pi r^2 h$

Surface Area (Lessons 7-7 and 7-8)

- Prism
Lateral Surface Area: $L = Ph$
Total Surface Area: $S = L + 2B$
- Pyramid
Lateral Surface Area: $L = P\ell$
Total Surface Area: $S = L + B$
- Cylinder
Lateral Surface Area: $L = 2\pi rh$
Total Surface Area: $S = L + 2B$

Similar Solids (Lesson 7-9)

- If two solids are similar with a scale factor of $\frac{a}{b}$, then the surface areas have a ratio of $\left(\frac{a}{b}\right)^2$ and the volumes have a ratio of $\left(\frac{a}{b}\right)^3$.

Key Vocabulary

base (p. 369)	lateral surface area (p. 386)
center (p. 352)	net (p. 385)
chord (p. 352)	pi (p. 352)
circle (p. 352)	polyhedron (p. 368)
circumference (p. 352)	prism (p. 368)
complex figure (p. 363)	pyramid (p. 369)
complex solid (p. 375)	radius (p. 352)
cone (p. 381)	regular pyramid (p. 393)
coplanar (p. 368)	similar solids (p. 399)
cylinder (p. 374)	slant height (p. 393)
diameter (p. 352)	solid (p. 368)
edge (p. 368)	total surface area (p. 386)
face (p. 368)	vertex (p. 368)
lateral face (p. 386)	volume (p. 373)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or
number to make a true sentence.

1. The flat surface of a prism is called a face.
2. Circumference is the distance around a circle.
3. The measure of the space occupied by a solid is called the total surface area.
4. A cylinder is a figure that has two parallel, congruent circular bases.
5. A solid is any two-dimensional figure.
6. The side of a prism is called a vertex.
7. The radius is the distance across a circle through its center.

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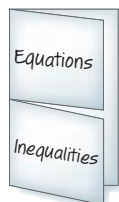


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Key Concepts

Algebraic Expressions (Lesson 8-1)

- Like terms contain the same variables to the same powers.
- A constant is a term without a variable.
- An algebraic expression is in simplest form if it has no like terms and no parentheses.

Equations (Lessons 8-2, 8-3, and 8-4)

- To solve a two-step equation, undo each operation in reverse order.
- To solve equations with variables on each side of the equals sign, use the Addition or Subtraction Property of Equality to write an equivalent equation with the variables on one side of the equals sign. Then solve the equation.

Inequalities (Lesson 8-6)

- When used to compare a variable and a number, inequalities can describe a range of values.

Inequality Properties (Lessons 8-7 and 8-8)

- When you add or subtract the same number from each side of an inequality, the inequality remains true.
- When you multiply or divide each side of an inequality by a positive number, the inequality remains true.
- When you multiply or divide each side of an inequality by a negative number, the direction of the symbol must be reversed for the inequality to be true.

Key Vocabulary

- coefficient (p. 417)
- constant (p. 417)
- equivalent expressions (p. 416)
- like terms (p. 417)
- simplest form (p. 418)
- simplifying the expression (p. 418)
- term (p. 417)
- two-step equation (p. 422)

Vocabulary Check

State whether each sentence is *true* or *false*. If *false*, replace the underlined word or number to make a true sentence.

1. Like terms are terms that contain different variables.
2. A two-step equation is an equation that contains two operations.
3. A coefficient is a term without a variable.
4. The numerical factor of a term that contains a variable is called the constant of the variable.
5. When plus or minus signs separate an algebraic expression into parts, each part is called a term.
6. An algebraic expression is in simplest form if it has no like terms and no parentheses.
7. The expressions $4(y + 7)$ and $4y + 28$ are two-step equations.
8. When you use the Distributive Property to combine like terms, you are simplifying the expression.

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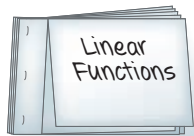


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Key Concepts

Functions (Lessons 9-1 and 9-2)

- A function is a relationship in which one value is dependent upon another.
- Functions can be represented by words, equations, tables, ordered pairs, and graphs.

Slope (Lesson 9-3)

- The slope m of a line passing through points (x_1, y_1) and (x_2, y_2) is the ratio of the difference in the y -coordinates to the corresponding difference in the x -coordinates.

Direct Variation (Lesson 9-4)

- A direct variation is a relationship in which the ratio of y to x is a constant, k .

Slope-Intercept Form (Lesson 9-5)

- An equation written in slope-intercept form is written as $y = mx + b$.
- When an equation is written in slope-intercept form, m is the slope and b is the y -intercept.

Systems of Equations (Lesson 9-6)

- Two equations together are called a system of equations.

Scatter Plots (Lesson 9-8)

- In a positive relationship, x increases and y increases.
- In a negative relationship, x increases and y decreases.
- In a no relationship, no obvious pattern exists between x and y .

Key Vocabulary

constant of variation (p. 483)	rise (p. 477)
direct variation (p. 483)	run (p. 477)
domain (p. 466)	scatter plot (p. 504)
function (p. 465)	slope (p. 477)
function table (p. 466)	slope-intercept form (p. 491)
linear function (p. 472)	system of equations (p. 498)
line of fit (p. 505)	y -intercept (p. 491)
range (p. 466)	

Vocabulary Check

Choose the correct term or number to complete each sentence.

1. The (domain, range) is the set of input values of a function.
2. The range is the set of (input, output) values of a function.
3. A relationship where one thing depends on another is called a (function, slope).
4. A (scatter plot, function table) is a graph that shows the relationship between two sets of data.
5. The (x -intercept, y -intercept) has the coordinates $(0, b)$.
6. The slope formula is $\left(\frac{y_2 - y_1}{x_2 - x_1}, \frac{x_2 - x_1}{y_2 - y_1}\right)$.
7. A line that is very close to most of the data points in a scatter plot is called a (line of fit, y -intercept).
8. The (rise, run) is the vertical change between two points on a line.
9. A(n) (dependent, independent) variable is the variable for the output of a function.

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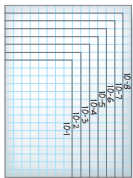


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Key Concepts

Functions (Lessons 10-1, 10-2, and 10-3)

- Linear functions have constant rates of change.
- Nonlinear functions do not have constant rates of change.
- Quadratic functions are functions in which the greatest power of the variable is 2.
- Cubic functions are functions in which the greatest power of the variable is 3.

Monomials (Lessons 10-5 through 10-8)

- To multiply powers with the same base, add their exponents.
- To divide powers with the same base, subtract their exponents.
- To find the power of a power, multiply the exponents.
- To find the power of a product, find the power of each factor and multiply.

Key Vocabulary

cube root (p. 554)

monomial (p. 539)

nonlinear function (p. 522)

quadratic function (p. 528)

Vocabulary Check

State whether each sentence is *true* or *false*. If *false*, replace the underlined word or number to make a true sentence.

1. The expression $y = x^2 - 3x$ is an example of a monomial.
2. A nonlinear function has a constant rate of change.
3. A quadratic function is a function whose greatest power is 2.
4. The product of $3x$ and $x^2 + 3x$ will have 3 terms.
5. A quadratic function is a nonlinear function.
6. The graph of a linear function is a curve.
7. To divide powers with the same base, subtract the exponents.
8. The Quotient of Powers states when dividing powers with the same base, subtract their exponents.
9. The graph of a cubic function is a straight line.

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Displays (Lessons 11-2, 11-3, and 11-6 to 11-8)

- Bar graphs show the number of items in specific categories.
- Box-and-whisker plots use a number line to show the distribution of a set of data.
- Circle graphs compare parts of the data to the whole.
- Histograms show the frequency of data that has been organized into equal intervals.
- Line graphs show change over a period of time.
- Line plots show how many times each number occurs in the data.
- Stem-and-leaf plots list all individual numerical data in a condensed form.

Measures of Central Tendency (Lesson 11-4)

- The mean of a set of data is the sum of the data divided by the number of items in a data set.
- The median of a set of data is the middle number of the ordered data, or the mean of the middle two numbers.
- The mode or modes of a set of data is the number or numbers that occur most often.

Measures of Variation (Lesson 11-5)

- The range of a set of data is the difference between the greatest and the least numbers in the set.
- The interquartile range is the range of the middle half of the data. It is the difference between the upper quartile and the lower quartile.

Key Vocabulary

box-and-whisker plot (p. 599)	measures of variation (p. 593)
circle graph (p. 576)	median (p. 585)
histogram (p. 570)	mode (p. 585)
interquartile range (p. 593)	outlier (p. 594)
lower quartile (p. 593)	quartiles (p. 593)
mean (p. 585)	range (p. 585)
measures of central tendency (p. 585)	stem-and-leaf plot (p. 606)
	upper quartile (p. 593)

Vocabulary Check

State whether each sentence is *true* or *false*.
If *false*, replace the underlined word or
number to make a true sentence.

1. A histogram is a bar graph that shows the frequency of data in intervals.
2. A variation is a piece of data that is more than 1.5 times the value of the interquartile range beyond the quartiles.
3. The range is one of the measures of central tendency.
4. The mean is the sum of the data divided by the number of pieces of data.
5. If you want to show how the parts compare to the whole, use a circle graph.
6. The mode is the middle number of a set of data.
7. The range is the difference between the greatest and the least values in a set of data.
8. The median is a data value that is quite separated from the rest of the data.

Study Guide and Review

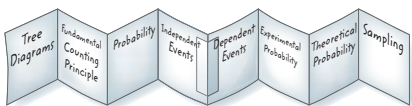


Download Vocabulary
Review from ca.g7math.com.

FOLDABLESTM Study Organizer

GET READY to Study

Be sure the following Key Concepts are noted in your Foldable.



Key Concepts

Counting Outcomes (Lesson 12-1)

- If event M can occur in m ways and is followed by event N that can occur in n ways, then the event M followed by the event N can occur in $m \cdot n$ ways.

Probability (Lessons 12-2 and 12-3)

- The probability of two independent events can be found by multiplying the probability of the first event by the probability of the second event.
- If two events, A and B , are dependent, then the probability of both events occurring is the product of the probability of A and the probability of B after A occurs.

Statistics (Lesson 12-5)

- An unbiased sample is representative of an entire population.
- A biased sample favors one or more parts of a population over others.

Key Vocabulary

biased sample (p. 648)	random (p. 627)
compound events (p. 631)	sample (p. 647)
convenience sample (p. 648)	sample space (p. 626)
dependent events (p. 632)	simple random sample (p. 647)
event (p. 626)	stratified random sample (p. 647)
experimental probability (p. 637)	systematic random sample (p. 647)
Fundamental Counting Principle (p. 627)	theoretical probability (p. 637)
independent events (p. 631)	tree diagram (p. 626)
outcome (p. 626)	unbiased sample (p. 647)
population (p. 647)	voluntary response sample (p. 648)
probability (p. 627)	

Vocabulary Check

Choose the correct term to complete each sentence.

1. A list of all possible outcomes is called the (sample space, event).
2. The (population, probability) of an event is the ratio of a specific outcome to the total number of outcomes.
3. A (combination, compound event) consists of two or more simple events.
4. For (independent, dependent) events, the outcome of one does not affect the other.
5. (Theoretical, Experimental) probability is based on known characteristics or facts.
6. A (simple random sample, convenience sample) is a biased sample.

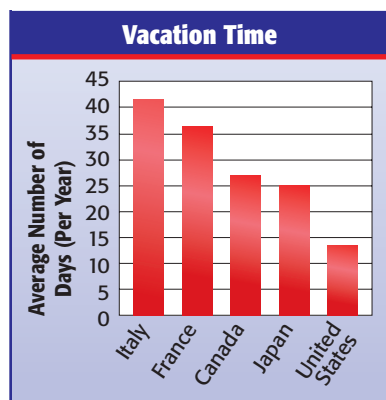
1 Displaying Data in Graphs



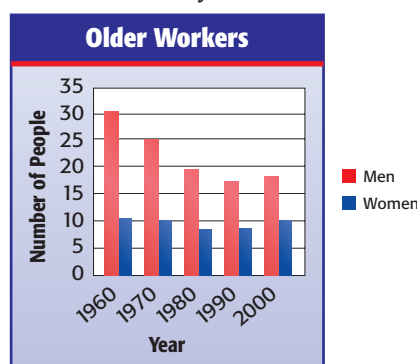
Reinforcement of **Standard 6NS2.3**
Analyze data displays . . .

Statistics involves collecting, analyzing, and presenting information, called **data**. Graphs display data to help readers make sense of the information.

- **Bar graphs** are used to compare the frequency of data. The bar graph below compares the average number of vacation days given by countries to their workers.
- **Double bar graphs** compare two sets of data. The double bar graph below shows the percent of men and women 65 and older who held jobs in various years.

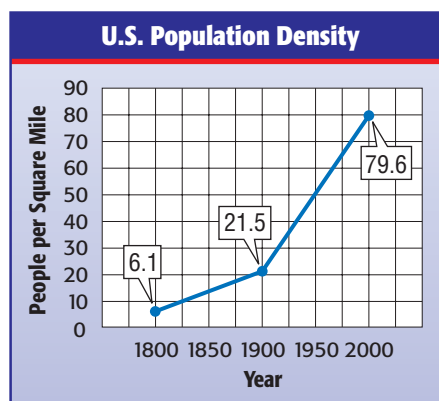


Source: The World Almanac

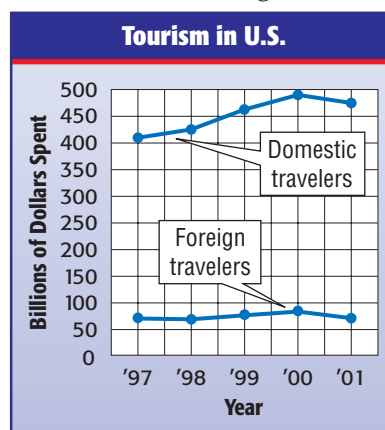


Source: The World Almanac

- **Line graphs** usually show how values change over time. The line graph below shows the number of people per square mile in the U.S. from 1800 through 2000.
- **Double line graphs**, like double bar graphs, show two sets of data. The double line graph below compares the amount of money spent by both domestic and foreign U.S. travelers.



Source: The World Almanac



Source: The World Almanac

- **Stem-and-leaf plots** are used to condense a set of data where the greatest place value of the data is used for the **stems** and the next greatest place value forms the **leaves**. Each data value can be seen in this type of graph.

The stem-and-leaf plot below contains this list of mathematics test scores:

95 76 64 88 93 68 99 96 74 75 92 80 76 85 91 70 62 81

The least number has 6 in the tens place.

The greatest number has 9 in the tens place.

The stems are 6, 7, 8, and 9.

The leaves are ordered from least to greatest.

Stem	Leaf
6	2 4 8
7	0 4 5 6 6
8	0 1 5 8
9	1 2 3 5 6 9 6 2 = 62

EXAMPLE**Choose a Display**

Shonny is writing a research paper about the Olympics for her social studies class. She wants to include a graph that shows how the times in the 400-meter run have changed over time. Should she use a line graph, bar graph, or stem-and-leaf plot?

Since the data would show how the times have changed over a period of time, she should choose a line graph.

Exercises

For Exercises 1–4, determine whether a bar graph, double bar graph, line graph, double line graph, or stem-and-leaf plot is the best way to display each of the following sets of data. Explain your reasoning.

1. how the income of households has changed from 1950 through 2000
2. the income of an average household in six different countries
3. the prices for a loaf of bread in twenty different supermarkets
4. the number of boys and the number of girls participating in six different school sports

For Exercises 5–9, refer to the bar graph, double bar graph, line graph, double line graph, and stem-and-leaf plot on page 662.

5. Write several sentences to describe the data shown in the graph titled “Vacation Time.” Include a comparison of the days worked for Canada and the U.S.
6. Write several sentences to describe the data shown in the graph titled “Older Workers.” What other type or types of graphs could you use to display this data? Explain your reasoning.
7. Write several sentences to describe the data shown in the graph titled “Tourism in U.S.” What other type or types of graphs could you use to display this data? Explain your reasoning.
8. Write several sentences to describe the data shown in the graph titled “U.S. Population Density.” What other type or types of graphs could you use to display this data? Explain your reasoning.
9. Write several sentences to describe the data shown in the stem-and-leaf plot of mathematics test scores. What is an advantage of showing the scores in this type of graph?

For Exercises 10–12, use the stem-and-leaf plot that shows the number of stories in the tallest buildings in Dallas, Texas.

10. How many buildings does the stem-and-leaf plot represent?
11. How many stories are there for the shortest building in the stem-and-leaf plot? the tallest building?
12. What is the median number of stories for these buildings? the mean number?

Stem	Leaf
2	7 9 9
3	0 1 1 1 3 3 4 4 6 6 7
4	0 2 2 5 9
5	0 0 0 0 2 5 6 8
6	0
7	2 2 7 = 27

2 Prime Factorization

When a whole number greater than 1 has *exactly* two factors, 1 and itself, it is called a **prime number**. When a whole number greater than 1 has more than two factors, it is called a **composite number**. The numbers 0 and 1 are *neither* prime *nor* composite. Notice that 0 has an endless number of factors and 1 has only one factor, itself.



Reinforcement of Standard 5NS1.4 Write prime factors of all numbers through 50 as the product of their prime factors by using exponents to show multiples of a factor (e.g., $24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$).

EXAMPLES Identify Numbers as Prime or Composite

Determine whether each number is *prime*, *composite*, or *neither*.

1 33

The numbers 1, 3, and 11 divide into 33 evenly. So, 33 is composite.

2 59

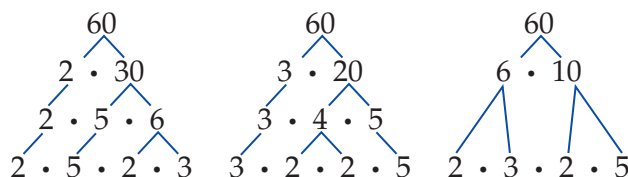
The only numbers that divide evenly into 59 are 1 and 59. So, 59 is prime.

When a number is expressed as a product of factors that are all prime, the expression is called the **prime factorization** of the number. A **factor tree** is useful in finding the prime factorization of a number.

EXAMPLE Write Prime Factorization

3 Use a factor tree to write the prime factorization of 60.

You can begin a factor tree for 60 in several ways.



Notice that the bottom row of “branches” in every factor tree is the same except for the order in which the factors are written. So, $60 = 2 \cdot 2 \cdot 3 \cdot 5$ or $2^2 \cdot 3 \cdot 5$.

Every number has a unique set of prime factors. This property of numbers is called the **Fundamental Theorem of Arithmetic**.

Exercises

Determine whether each number is *prime*, *composite*, or *neither*.

- | | | | |
|--------|---------|---------|---------|
| 1. 45 | 2. 23 | 3. 1 | 4. 13 |
| 5. 27 | 6. 96 | 7. 37 | 8. 0 |
| 9. 177 | 10. 233 | 11. 507 | 12. 511 |

Write the prime factorization of each number.

- | | | | |
|--------|---------|---------|-----------|
| 13. 20 | 14. 49 | 15. 225 | 16. 32 |
| 17. 25 | 18. 36 | 19. 51 | 20. 75 |
| 21. 80 | 22. 117 | 23. 72 | 24. 4,900 |

3 Greatest Common Factor

The greatest of the factors common to two or more numbers is called the **greatest common factor (GCF)** of the numbers. One way to find the GCF is to list the factors of the numbers.

EXAMPLE Find the GCF

- 1 Find the greatest common factor of 36 and 60.

METHOD 1 List the factors.

factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

The greatest common factor of 36 and 60 is 12.

Common factors of 36 and 60: 1, 2, 3, 4, 6, 12

METHOD 2 Use prime factorization.

$$36 = 2 \cdot 2 \cdot 3 \cdot 3$$

$$60 = 2 \cdot 2 \cdot 3 \cdot 5$$

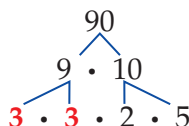
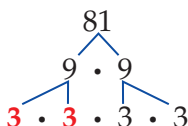
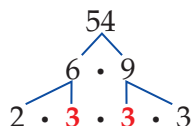
The GCF is $2 \cdot 2 \cdot 3$ or 12.

Common prime factors of 36 and 60: 2, 2, 3

EXAMPLE Find the GCF

- 2 Find the greatest common factor of 54, 81, and 90.

Use a factor tree to find the prime factorization of each number.



The common prime factors of 54, 81, and 90 are 3 and 3.

The GCF of 54, 81, and 90 is $3 \cdot 3$ or 9.

Exercises

Find the GCF of each set of numbers.

- | | | | |
|------------|-----------|----------------|----------------|
| 1. 45, 20 | 2. 27, 54 | 3. 24, 48 | 4. 63, 84 |
| 5. 40, 60 | 6. 32, 48 | 7. 30, 42 | 8. 54, 72 |
| 9. 36, 144 | 10. 3, 51 | 11. 24, 36, 42 | 12. 35, 49, 84 |

13. **DESIGN** Suppose you are tiling a tabletop with 6-inch square tiles. How many of these squares will be needed to cover a 30-inch by 24-inch table?

Two or more numbers are **relatively prime** if their greatest common factor is 1. Determine whether each set of numbers is relatively prime.

- | | | | |
|-----------|-----------|-----------|---------------|
| 14. 9, 19 | 15. 7, 21 | 16. 3, 51 | 17. 4, 28, 31 |
|-----------|-----------|-----------|---------------|

4 Simplifying Fractions

Fractions, mixed numbers, decimals, and integers are examples of **rational numbers**. When a rational number is represented as a fraction, it is often expressed in **simplest form**. A fraction is in simplest form when the GCF of the numerator and denominator is 1.



Reinforcement of Standard 6NS2.4 Determine the least common multiple and the greatest common divisor of whole numbers use them to solve problems with fractions . . .

EXAMPLE Simplify Fractions

- 1 Write $\frac{30}{45}$ in simplest form.

METHOD 1 Divide by the GCF.

$$\begin{array}{ll} 30 = 2 \cdot \boxed{3} \cdot \boxed{5} & \text{Factor the numerator.} \\ 45 = 3 \cdot \boxed{3} \cdot \boxed{5} & \text{Factor the denominator.} \end{array}$$

The GCF of 30 and 45 is $3 \cdot 5$ or 15.

$$\begin{aligned} \frac{30}{45} &= \frac{30 \div 15}{45 \div 15} && \text{Divide the numerator and denominator by the GCF, 15.} \\ &= \frac{2}{3} \end{aligned}$$

METHOD 2 Use prime factorization.

$$\begin{aligned} \frac{30}{45} &= \frac{2 \cdot 3 \cdot 5}{3 \cdot 3 \cdot 5} && \text{Write the prime factorization of the numerator and denominator.} \\ &= \frac{2 \cdot \overset{1}{\cancel{3}} \cdot \overset{1}{\cancel{5}}}{\overset{1}{\cancel{3}} \cdot 3 \cdot \overset{1}{\cancel{5}}} && \text{Divide the numerator and denominator by the GCF, } 3 \cdot 5. \\ &= \frac{2}{3} && \text{Simplify.} \end{aligned}$$

Exercises

Write each fraction in simplest form. If the fraction is already in simplest form, write *simplest form*.

- | | | | | |
|----------------------|------------------------|------------------------|-----------------------|-------------------------|
| 1. $\frac{8}{72}$ | 2. $\frac{27}{45}$ | 3. $\frac{60}{75}$ | 4. $\frac{36}{54}$ | 5. $\frac{3}{9}$ |
| 6. $\frac{15}{25}$ | 7. $\frac{36}{81}$ | 8. $\frac{18}{54}$ | 9. $\frac{14}{66}$ | 10. $\frac{24}{54}$ |
| 11. $\frac{15}{24}$ | 12. $\frac{48}{72}$ | 13. $\frac{24}{120}$ | 14. $\frac{66}{88}$ | 15. $\frac{72}{98}$ |
| 16. $\frac{45}{100}$ | 17. $\frac{7}{91}$ | 18. $\frac{15}{100}$ | 19. $\frac{15}{60}$ | 20. $\frac{17}{51}$ |
| 21. $\frac{6}{9}$ | 22. $\frac{16}{40}$ | 23. $\frac{6}{16}$ | 24. $\frac{64}{68}$ | 25. $\frac{30}{80}$ |
| 26. $\frac{2}{15}$ | 27. $\frac{90}{120}$ | 28. $\frac{75}{89}$ | 29. $\frac{16}{96}$ | 30. $\frac{133}{140}$ |
| 31. $\frac{99}{300}$ | 32. $\frac{50}{1,000}$ | 33. $\frac{90}{6,000}$ | 34. $\frac{150}{400}$ | 35. $\frac{10}{10,000}$ |

36. Both the numerator and the denominator of a fraction are even. Can you tell whether the fraction is in simplest form? Explain.
37. **WEATHER** The rainiest place on Earth is Waialeale, Hawaii. Of 365 days per year, the average number of rainy days is 335. Write a fraction in simplest form to represent these rainy days as a part of a year.
38. **OLYMPICS** In the 2000 Olympics, Brooke Bennett of the U.S. swam the 800-meter freestyle event in about 8 minutes. Express 8 minutes in terms of hours using a fraction in simplest form.

6NS2.4 Determine the least common multiple and the greatest common divisor of whole numbers ...

5 Least Common Multiple

A **multiple** of a number is the product of that number and any whole number.

EXAMPLE List Multiples

- 1** List the first six multiples of 15.

$$0 \cdot 15 = 0, 1 \cdot 15 = 15, 2 \cdot 15 = 30, 3 \cdot 15 = 45, 4 \cdot 15 = 60, 5 \cdot 15 = 75$$

The first six multiples of 15 are 0, 15, 30, 45, 60, 75.

The least of the nonzero common multiples of two or more numbers is called the **least common multiple (LCM)** of the numbers. To find the LCM of two or more numbers, you can list the multiples of each number until a common multiple is found, or you can use prime factorization.

EXAMPLE Find the LCM

- 2** Find the LCM of 12 and 18.

METHOD 1 List the multiples.

multiples of 12: 0, 12, 24, **36**, 48, ...

multiples of 18: 0, 18, **36**, 54, 72, ...

The LCM of 12 and 18 is 36.

Remember that the LCM is a *nonzero* number.

METHOD 2 Use prime factorization.

$$\begin{array}{l} 12 = 2 \cdot 2 \cdot 3 \\ 18 = 2 \cdot 3 \cdot 3 \end{array}$$

Write the prime factorization of each number.

Multiply the factors, using the common factors only once.

The LCM is $2 \cdot 2 \cdot 3 \cdot 3$ or 36.

Exercises

List the first six multiples of each number.

- | | | | | |
|-------|--------|------|------|-------|
| 1. 7 | 2. 11 | 3. 4 | 4. 5 | 5. 14 |
| 6. 25 | 7. 150 | 8. 2 | 9. 3 | 10. 6 |

Find the least common multiple (LCM) of each set of numbers.

- | | | | |
|---------------|---------------|----------------|-------------------|
| 11. 8, 20 | 12. 15, 18 | 13. 12, 16 | 14. 7, 12 |
| 15. 20, 50 | 16. 16, 24 | 17. 2, 7, 8 | 18. 2, 3, 5 |
| 19. 4, 8, 12 | 20. 7, 21, 5 | 21. 8, 28, 30 | 22. 10, 12, 14 |
| 23. 24, 12, 6 | 24. 45, 10, 6 | 25. 2, 3, 5, 7 | 26. 2, 15, 25, 36 |

27. **CIVICS** In the United States, a president is elected every four years. Members of the House of Representatives are elected every two years. Senators are elected every six years. If a voter had the opportunity to vote for a president, a representative, and a senator in 1996, what will be the next year the voter has a chance to make a choice for a president, a representative, and the same Senate seat?

6 Plotting Points on a Coordinate Plane

An ordered pair of numbers is used to locate any point on a coordinate plane. The first number is called the x -coordinate. The second number is called the y -coordinate.



Reinforcement of Standard 5AF1.4
Identify and graph ordered pairs in the four quadrants of the coordinate plane.

x -coordinate y -coordinate
 $(-4, -3)$
 ordered pair

EXAMPLES Identify Ordered Pairs

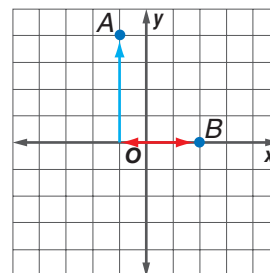
1 Write the ordered pair that names point A.

Step 1 Start at the origin.

Step 2 Move left on the x -axis to find the x -coordinate of point A, which is -1 .

Step 3 Move up along the y -axis to find the y -coordinate which is 4.

The ordered pair for point A is $(-1, 4)$.



2 Write the ordered pair that names point B.

The x -coordinate of B is 2. Since the point lies on the x -axis, its y -coordinate is 0.

The ordered pair for point B is $(2, 0)$.

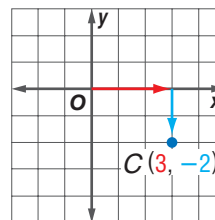
EXAMPLE Graph an Ordered Pair

3 Graph and label the point C(3, -2) on a coordinate plane.

Step 1 Start at the origin.

Step 2 Since the x -coordinate is 3, move 3 units right.

Step 3 Since the y -coordinate is -2 , move down 2 units. Draw and label a dot.



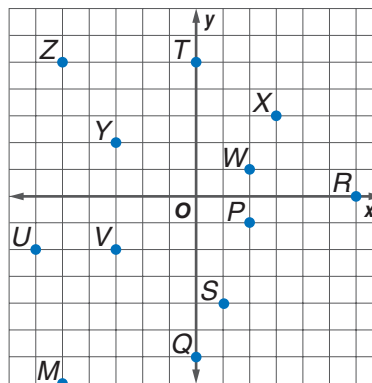
Exercises

Name the ordered pair for the coordinates of each point on the coordinate plane.

- | | | |
|-------|-------|-------|
| 1. Z | 2. X | 3. W |
| 4. Y | 5. T | 6. V |
| 7. U | 8. S | 9. Q |
| 10. R | 11. P | 12. M |

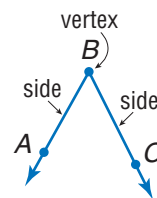
Graph each point on the same coordinate plane.

- | | | |
|---------------|---------------|---------------|
| 13. A(4, 7) | 14. C(1, 0) | 15. B(0, 7) |
| 16. E(-1, -2) | 17. D(-4, -7) | 18. F(-10, 3) |
| 19. G(9, 9) | 20. J(7, -8) | 21. K(-6, 0) |
| 22. H(0, -3) | 23. I(4, 0) | 24. M(2, 7) |
| 25. N(8, -1) | 26. L(-1, -1) | 27. P(3, 3) |



7 Measuring and Drawing Angles

Two rays that have a common endpoint form an **angle**. The common endpoint is called the **vertex**, and the two rays that make up the angle are called the **sides** of the angle. A circle can be divided into 360 equal sections. Each section is one **degree**. You can use a **protractor** to measure an angle in degrees and draw an angle with a given degree measure.



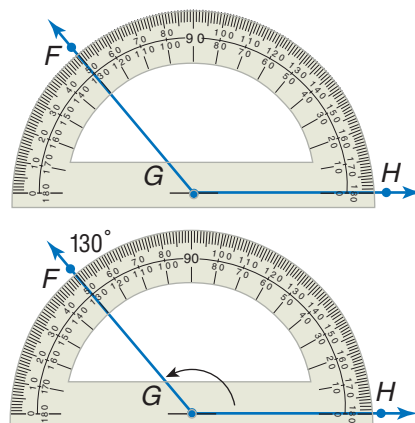
EXAMPLE Measure an Angle

1 Use a protractor to measure $\angle FGH$.

Step 1 Place the center point of the protractor's base on vertex G . Align the straight side with side $\overrightarrow{GH}$ so that the marker for 0° is on the ray.

Step 2 Use the scale that begins with 0° at $\overrightarrow{GH}$. Read where the other side of the angle, $\overrightarrow{GF}$, crosses this scale.

The measure of angle FGH is 130° .
 Using symbols, $m\angle FGH = 130^\circ$.



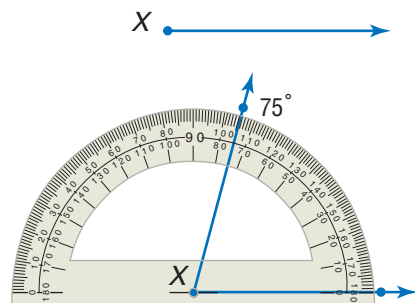
EXAMPLE Draw an Angle

2 Draw $\angle X$ having a measure of 75° .

Step 1 Draw a ray. Label the endpoint X .

Step 2 Place the center point of the protractor's base on point X . Align the mark labeled 0 with the ray.

Step 3 Use the scale that begins with 0. Locate the mark labeled 75. Then draw the other side of the angle.



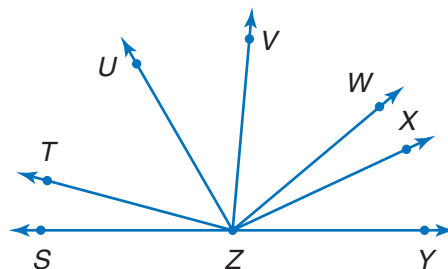
Exercises

Use a protractor to find the measure of each angle.

- | | | |
|-----------------|-----------------|-----------------|
| 1. $\angle XZY$ | 2. $\angle SZT$ | 3. $\angle SZY$ |
| 4. $\angle UZX$ | 5. $\angle TZW$ | 6. $\angle UZV$ |

Use a protractor to draw an angle having each measurement.

- | | | |
|-----------------|-----------------|-----------------|
| 7. 40° | 8. 70° | 9. 65° |
| 10. 110° | 11. 85° | 12. 90° |
| 13. 155° | 14. 140° | 15. 117° |

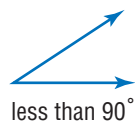


8 Classifying Angles

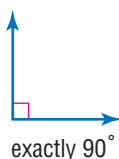
Angles are most often classified by their measures.

- **Acute angles** have measures less than 90° .
- **Right angles** have measures equal to 90° .
- **Obtuse angles** have measures between 90° and 180° .
- **Straight angles** have measures equal to 180° .

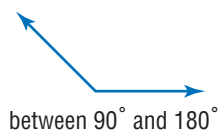
Acute angle



Right angle



Obtuse angle



Straight angle



Reinforcement of

Standard

5MG2.1 Measure, identify, and draw angles, perpendicular and parallel lines, and triangles by using appropriate tools . . .

EXAMPLES Classify Angles

Classify each angle as *acute*, *obtuse*, *right*, or *straight*.

1



$m\angle 1$ is greater than 90° .
So, $\angle 1$ is an obtuse angle.

2

$\angle C = 75^\circ$

$\angle C$ is less than 90° .
So, $\angle C$ is an acute angle.

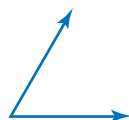
Exercises

Classify each angle as *acute*, *obtuse*, *right*, or *straight*.

1.



2.



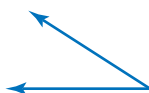
3.



4.



5.



6.



7.



8.



9.



10. $\angle F = 150^\circ$

11. $\angle J = 75^\circ$

12. $\angle P = 89^\circ$

13. $\angle C = 90^\circ$

14. $\angle K = 180^\circ$

15. $\angle A = 177^\circ$

16. $\angle M = 45^\circ$

17. $\angle B = 91^\circ$

18. $\angle D = 142^\circ$

Reinforcement of Standard
6MG2.2 Use the properties of complementary and supplementary angles and the sum of the angles of a triangle to solve problems involving an unknown angle.

9 Triangles

A **triangle** is a figure formed by three line segments that intersect only at their end points. The sum of the measures of angles of a triangle is 180° .

EXAMPLE Find a Missing Angle Measurement

1 Find the value of x in $\triangle RST$.

$$m\angle R + m\angle S + m\angle T = 180$$

$$x + 72 + 74 = 180$$

$$x + 146 = 180$$

$$\underline{- 146 = - 146}$$

$$x = 34$$

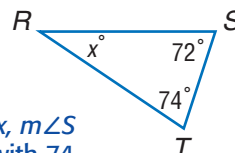
The value of x is 34.

The sum of the measures is 180.

Replace $m\angle R$ with x , $m\angle S$ with 72, and $m\angle T$ with 74.

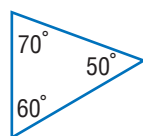
Simplify.

Subtract 146 from each side.



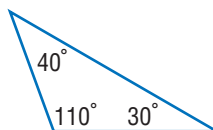
All triangles have at least two acute angles and can be classified by the measure of the third angle.

Acute Triangle



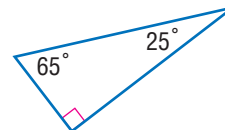
three acute angles

Obtuse Triangle



one obtuse angle

Right Triangle



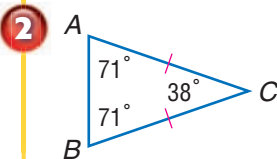
one right angle

Triangles can also be classified by the number of congruent sides.

- **Scalene triangles** have no congruent sides
- **Isosceles triangles** have at least two sides congruent
- **Equilateral triangles** have 3 congruent sides

EXAMPLE Classify Triangles

Classify each triangle by its angles and by its sides.



Angles $\triangle ABC$ has all acute angles.

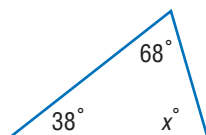
Sides $\triangle ABC$ has two congruent sides.

So, $\triangle ABC$ is an acute isosceles triangle.

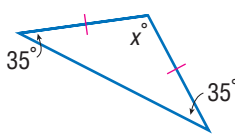
Exercises

Find the value of x in each triangle. Then, classify each triangle by its angles and by its sides.

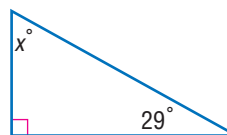
1.



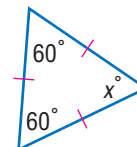
2.



3.



4.

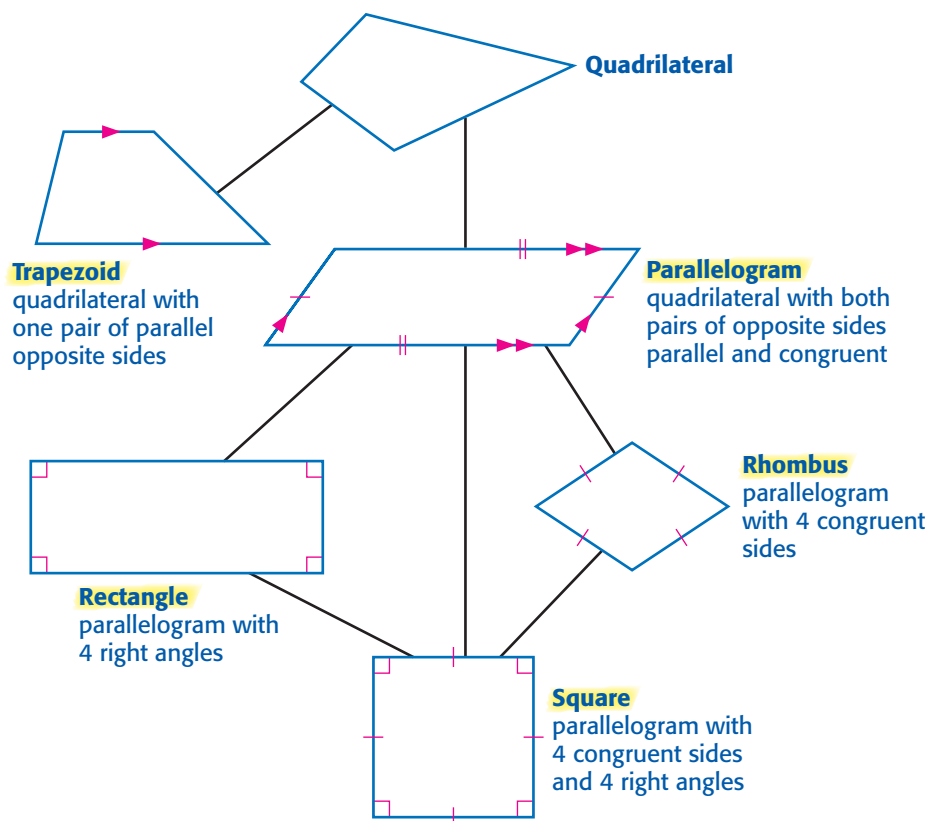


10 Classifying Quadrilaterals

A **quadrilateral** is a polygon that has four sides and four angles. The concept map below shows how quadrilaterals are classified. Notice that the diagram goes from the most general type of quadrilateral to most specific.



Reinforcement of Standard
5MG2.1 Measure, identify, and draw angles, perpendicular and parallel lines, ...



EXAMPLES Classifying Quadrilaterals

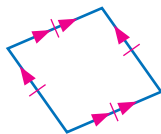
Classify each quadrilateral using name that *best* describes it.

1



The quadrilateral has one pair of parallel sides. It is a trapezoid.

2

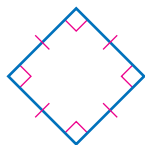


The quadrilateral is a parallelogram with four congruent sides. It is a rhombus.

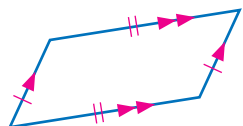
Exercises

Classify each quadrilateral using the best name that describes it.

1.



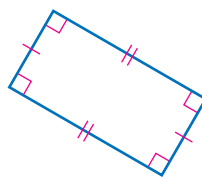
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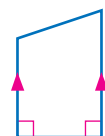
3.



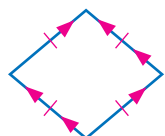
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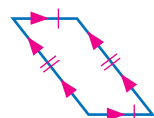
5.



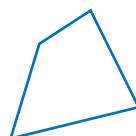
6.



7.

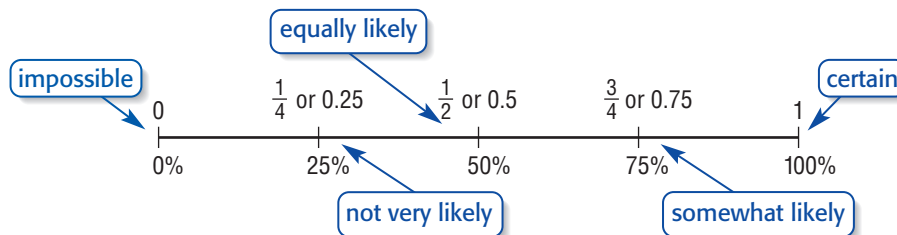


8.



11 Probability of Simple Events

A **simple event** is a specific outcome or type of outcome. **Probability** is the chance that something will happen. The probability of an event is a ratio that compares the number of favorable outcomes to the number of possible outcomes. The probability that an event will happen is between 1 and 0. A probability can be expressed as a fraction, a decimal, or a percent.



EXAMPLE Find Probabilities

- 1** A box contains 5 green pens, 3 blue pens, 8 black pens, and 4 red pens. A pen is picked at random. What is the probability the pen is green?

There are $5 + 3 + 8 + 4$ or 20 pens in the box.

$$P(\text{green}) = \frac{\text{green pens}}{\text{total number of pens}} \quad \text{Definition of probability}$$

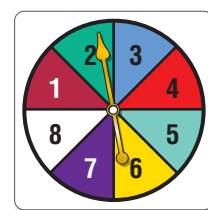
$$= \frac{5}{20} \text{ or } \frac{1}{4} \quad \text{There are 5 green pens out of 20 pens.}$$

The probability the pen is green is $\frac{1}{4}$. The probability can also be written as 0.25 or 25%.

Exercises

The spinner is used for a game. Write each probability as a fraction, a decimal, and a percent.

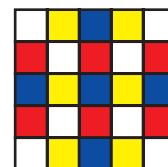
1. $P(5)$
2. $P(\text{even})$
3. $P(\text{greater than } 5)$
4. $P(\text{not } 2)$
5. $P(\text{an integer})$
6. $P(\text{less than } 7)$



7. **COINS** A jar has 11 nickels, 28 dimes, 39 pennies, 22 quarters, and 8 silver dollars. What is the probability that the first coin picked is a silver dollar?

A beanbag is tossed on the square at the right. It lands at random in a small square. Write each probability as a fraction, a decimal, and a percent.

8. $P(\text{red})$
9. $P(\text{blue})$
10. $P(\text{white or yellow})$
11. $P(\text{blue or red})$
12. $P(\text{not green})$
13. $P(\text{brown})$



14. **MONTHS** What is the probability that a month picked at random starts with the letter J?

12 Perimeter and Area of Rectangles

The distance around a geometric figure is called its **perimeter**. The perimeter P of a rectangle is twice the sum of the length ℓ and width w , or $P = 2\ell + 2w$. The measure of the surface enclosed by a figure is its **area**. The area A of a rectangle is the product of the length ℓ and width w , or $A = \ell w$.



Reinforcement of Standard 6AF3.1 Use variables in expressions describing geometric quantities (e.g., $P = 2w + 2x, \dots$)

EXAMPLES Find the Perimeter and Area of a Rectangle

1 Find the perimeter of the rectangle.

$$P = 2\ell + 2w$$

Write the formula.

$$P = 2(27) + 2(12)$$

Replace ℓ with 27 and w with 12.

$$P = 54 + 24$$

Multiply.

$$P = 78$$

Add.

The perimeter is 78 feet.



2 Find the area of the rectangle.

$$A = \ell w$$

Write the formula.

$$A = 27 \cdot 12$$

Replace ℓ with 27 and w with 12.

$$A = 324$$

Multiply.

The area is 324 square feet.

A square is a rectangle with sides that are all the same length. The perimeter P of a square is four times the side length s , or $P = 4s$. Its area A is the square of the side length, or $A = s^2$.

EXAMPLE Find the Perimeter and Area of a Square

3 Find the perimeter and area of a square with side length 6.4 meters.

$$P = 4s$$

Write the formula.

$$P = 4(6.4)$$

Replace s with 6.4.

$$P = 25.6$$

Multiply.

The perimeter is 25.6 meters.

$$A = s^2$$

Write the formula.

$$A = (6.4)^2$$

Replace s with 6.4.

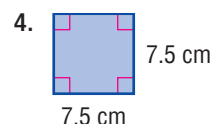
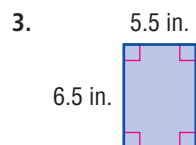
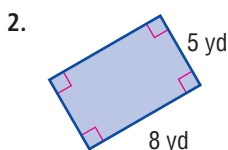
$$A = 40.96$$

$6.4^2 = 6.4 \cdot 6.4$ or 40.96

The area is 40.96 square meters.

Exercises

Find the perimeter and area of each figure.



5. rectangle: 3 mm by 5 mm

7. square: side length, 75 ft

9. rectangle: 4.3 cm by 2.7 cm

11. square: side length of 87 km

6. rectangle: 144 mi by 25 mi

8. square: side length, 0.75 yd

10. square: side length of 625 m

12. rectangle: 875.5 mm by 245.3 mm



Reinforcement of Standard

6AF3.1 Use variables in expressions describing geometric quantities . . .

13 Area of Parallelograms, Triangles, and Trapezoids

The areas of parallelograms, triangles, and trapezoids can be found using the formulas found in the table below.

Shape	Formula	Model
Parallelogram	$A = bh$	
Triangle	$A = \frac{1}{2}bh$	
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	

EXAMPLE

Find the Area of a Trapezoid

1 Find the area of the trapezoid.

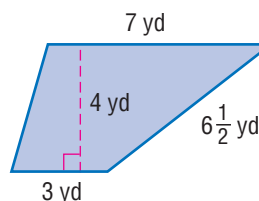
The height is 4 yards. The lengths of the bases are 7 yards and 3 yards.

$$A = \frac{1}{2}h(b_1 + b_2) \quad \text{Area of a trapezoid.}$$

$$A = \frac{1}{2}(4)(7 + 3) \quad \text{Replace } h \text{ with 4, } b_1 \text{ with 7, and } b_2 \text{ with 3.}$$

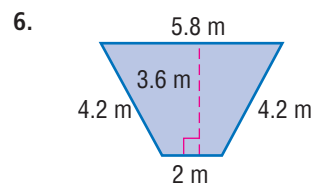
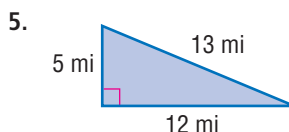
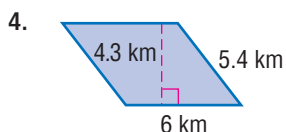
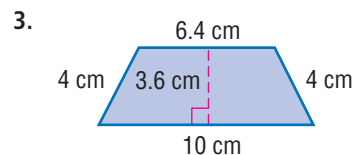
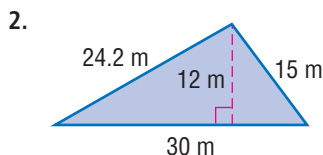
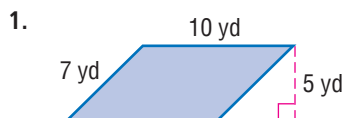
$$A = \frac{1}{2}(4)(10) \text{ or } 20 \quad \text{Simplify.}$$

The area of the trapezoid is 20 square yards.



Exercises

Find the area of each figure.



Practice by Standard: Number Sense

Standard Set 1.0: Know the properties of, and compute with, rational numbers expressed in a variety of forms.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 Which of the following is equivalent to $\left(\frac{3}{4}\right)^3$? (7NS1.2)
- A $\frac{1}{256}$
 - B $\frac{4}{27}$
 - C $\frac{27}{64}$
 - D $\frac{9}{16}$

- 2 Patrick works in a clothing store. His monthly sales are listed in the table below.

Month	Sales
January	\$1560
February	\$4075
March	\$3050

If his commission is 15%, to the nearest dollar how much commission did he make in January? (7NS1.7)

- F \$1,003 H \$234
- G \$311 J \$146

- 3 Which of the following is equivalent to $\frac{9}{4}$? (7NS1.3)
- A 0.444
 - B 1.8
 - C 2.25
 - D 2.75

QUICK Review

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

To find the value of the expression, use $\frac{3}{4}$ as a factor three times.

For more help with raising rational numbers to whole number powers, see page 126.

READING HINT Commission is found by multiplying the amount of sales by the percent of commission.

First, locate the amount of sales for January. Then multiply this by 0.15.

For more help with commission, see page 279.

STRATEGY As you examine each choice, put a check beside the ones that AREN'T equivalent. The one without a check is the correct answer.

To write a fraction as a decimal, divide the numerator by the denominator.

For more help with dividing integers, see page 85.

QUICK Practice

- 4 Which shows 9,452 written in scientific notation? (7NS1.1)

F 9.452×10^1

G 9.452×10^2

H 9.452×10^3

J 9.452×10^4

- 5 The following table gives the prices of several items at Brenda's Boutique.

Item	Price
T-shirt	\$20
Jeans	\$35
Necklaces	\$15

Brenda is having a sale and marking everything 20% off. How much will a pair of jeans cost on sale? (7NS1.7)

A \$7

C \$28

B \$21

D \$30

- 6 Pearl borrowed \$500 at 15% simple interest for one year. If she makes no payments that year, how much interest will she owe at the end of the year? (7NS1.7)

F \$50

H \$100

G \$75

J \$575

- 7 Five friends went out to eat together. The total bill was \$46.82 before tax. The tax was 6% of the bill. If they all split the bill including tax evenly, which expression can be used to find the amount of tax that each person paid? (7NS1.7)

A $(46.82 + 0.06) \div 5$

B $(46.82 \cdot 0.06) + 5$

C $(46.82 \div 0.06) \cdot 5$

D $(46.82 \cdot 0.06) \div 5$

QUICK Review

STRATEGY Review definitions of key terms.

In scientific notation, the exponent for the 10 is the same as the number of places the decimal in the other factor needs to move to get your original number.

For more help with scientific notation, see page 130.

STRATEGY Sometimes tables will give more information than you will use. In this table, you won't use the price of the T-shirt or the necklace.

Since the discount is 20%, the sale price is 80% of the original amount. Multiply the original price of the jeans by 0.80 to get the sale price.

For more help with discounts, see page 287.

STRATEGY Eliminate answers that aren't reasonable. She can't owe more interest than the amount that she borrowed.

To find the amount of interest she will owe, multiply the amount that she borrowed by 0.15.

For more help with simple interest, see page 290.

STRATEGY How do you determine tax due for a meal?

Begin by finding the amount of tax due on the bill and then the amount of tax per person.

For more help with using the percent equation, see page 279.

Practice by Standard: Number Sense

Standard Set 2.0: Students use exponents, powers, and roots, and use exponents in working with fractions.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 Which of the following has the same value as $6^{-4} \times 6^6$? (7NS2.1)
- A 6^{-24}
 - B 6^{-10}
 - C 6^2
 - D 6^{10}

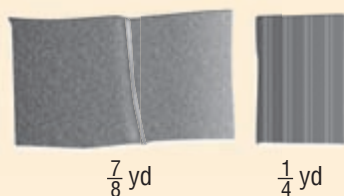
QUICK Review

STRATEGY Positive exponents represent repeated multiplication; negative exponents represent repeated division.

Since the bases are the same, add the exponents when multiplying.

For more help with multiplying integers with exponents, see page 127.

- 2 Latisha has two pieces of fabric as shown below.



What is the sum of the lengths of fabric? (7NS2.2)

- F $\frac{2}{3}$ yard
- G $\frac{8}{9}$ yard
- H $\frac{9}{8}$ yard
- J $\frac{3}{2}$ yard

READING HINT Sum means addition.

You can find the sum by adding the length of the first piece of fabric to the length of the second piece of fabric.

For more help with adding fractions, see page 114.

- 3 Which expression is equivalent to $\frac{7^5}{7^3}$? (7NS2.3)
- A 7^{15}
 - B 7^8
 - C 7^2
 - D $7^{\frac{5}{3}}$

STRATEGY Work the problem in two different ways to check your answer.

To divide rational numbers with exponents, check to make sure the base is the same. If so, keep the base and subtract exponents.

For more help with dividing rational numbers with exponents, see page 544.

QUICK Practice

- 4 Which of the following shows the next step using the least common denominator to simplify $\frac{1}{3} + \frac{3}{5}$?

(7NS2.2)

- F $\left(\frac{1}{3} \times \frac{5}{5}\right) + \left(\frac{3}{5} \times \frac{3}{3}\right)$
 G $\left(\frac{1}{3} \times \frac{1}{5}\right) + \left(\frac{3}{5} \times \frac{1}{3}\right)$
 H $\left(\frac{1}{3} \times \frac{3}{3}\right) + \left(\frac{3}{5} \times \frac{5}{5}\right)$
 J $\left(\frac{1}{3} \times \frac{1}{3}\right) + \left(\frac{3}{5} \times \frac{1}{5}\right)$

- 5 Between which two integers does the square the square root of 56 lie?

(7NS2.4)

- A 6 and 7
 B 7 and 8
 C 8 and 9
 D 55 and 56

- 6 Which expression is equivalent to $3^3\left(\frac{1}{3^2}\right)$? (7NS2.3)

- F 3
 G 3^2
 H 3^5
 J 3^6

- 7 Mrs. Jennings wrote four irrational numbers on the board and asked Marcy to choose the number closest to 5. Which irrational number should she choose? (7NS2.4)

- A $\sqrt{6}$
 B $\sqrt{10}$
 C $\sqrt{24}$
 D $\sqrt{33}$

QUICK Review

STRATEGY Eliminate any choices in which the original fractions are *not* multiplied by a fraction that is equivalent to 1.

To find least common denominator, list the multiples of 5. The first multiple that is also a multiple of 3 is the least common denominator.

For more help with adding fractions, see page 114.

READING HINT The square root of a number is the number that when multiplied by itself will give you the original number.

Find the perfect squares that are greater and less than 56 and as close to 56 as possible.

For more help with square roots, see page 148.

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

Write $\frac{1}{3^2}$ as 3^{-2} and multiply.

For more help with computing with powers, see page 544.

READING HINT Irrational numbers are nonrepeating, nonterminating decimals.

Find the square root that is closest to $\sqrt{25}$, or 5.

For more help with irrational numbers, see page 155.

Practice by Standard: Algebra and Functions

Standard Set 1.0: Students express quantitative relationships by using algebraic terminology, expressions, equations, inequalities, and graphs.

DIRECTIONS

Choose the correct answer.

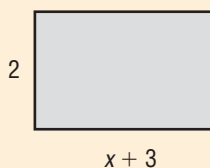
QUICK Practice

- Which property is shown in the equation below? (7AF1.3)
 $12 + x + 16 = 12 + 16 + x$
 A Associative Property of Addition
 B Commutative Property of Addition
 C Distributive Property
 D Reflexive Property

- The sum of a number (x) and 12 is 54. Which equation shows this relationship? (7AF1.1)
 F $54x = 12$
 G $12 + x = 54$
 H $12 - x = 54$
 J $54 + x = 12$

- Which expression is equivalent to $x(2 + y)$? (7AF1.3)
 A $2x + y$ C $2x + 2y$
 B $2xy$ D $2x + xy$

- What is the area, in square units of the rectangle below? (7AF1.3)



- F $2x + 3$ H $2x + 6$
 G $x + 5$ J $2x + 10$

QUICK Review

STRATEGY Try to eliminate some of the answer choices by recalling the definitions.

Notice that the order of the addends in the expression was changed.

For more help with properties of addition, see page 31.

READING HINT Sum means addition.

You can answer this question by first writing the sum of a number and 12 as an addition statement. Then, set this equal to 54.

For more help with writing algebraic equations, see page 57.

STRATEGY Replace x with one value and y with another in each expression. Evaluate to find the choice with the same value as $x(2 + y)$.

Use the Distributive Property and the Commutative Property of Multiplication.

For more help with the Distributive Property, see page 31.

READING HINT The area of a rectangle is found by multiplying the length by the width.

Replace w with 2 and ℓ with $(x + 3)$ in the equation $A = \ell w$. Then use the Distributive Property.

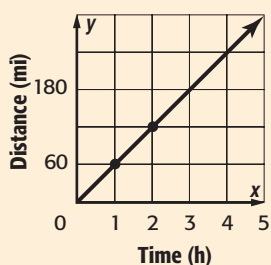
For more help with the Distributive Property, see page 31.

QUICK Practice

- 5 If $a = 2$ and $b = 5$, then $a - 2b =$
(7AF1.2)

A -8 C 1
B 0 D 8

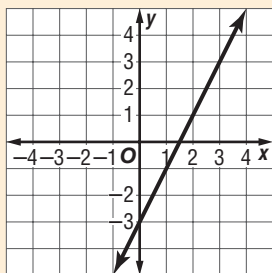
- 6 The following graph shows the distance Henry drove, on average, as he went on vacation.



Which statement is *not* true?
(7AF1.5)

- F He drove 120 miles in 2 hours.
G His average rate varied.
H His average rate was a constant.
J He drove at a rate of 60 miles per hour.

- 7 Which of the following points lies on the line graphed? (7AF1.5)



- A $W(-1, -1)$ C $Y(3, 3)$
B $X(-2, 1)$ D $Z(0, 1.5)$

QUICK Review

STRATEGY Make sure to record your work. If you reach an answer that is not there, you can find your error.

Replace a with 2 and b with 5 in the expression.

For more help with evaluating algebraic expressions, see page 30.

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

To answer the question, look at the coordinates $(2, 120)$ and what each represents by looking at the labels on the grid.

For more help with analyzing ordered pairs, see page 471.

STRATEGY How do you know which point is on the line?

Graph each point to identify which is on the line.

For more help with coordinate grids, see page 668.

Practice by Standard: Algebra and Functions

Standard Set 2.0: Students interpret and evaluate expressions involving integer powers and simple roots.

DIRECTIONS

Choose the correct answer.

QUICK Practice

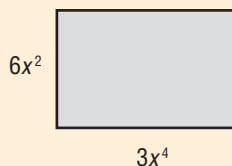
- 1 Which expression has the same value as a^4 ? (7AF2.1)

A $a \cdot a \cdot a$
 B $a \cdot a \cdot a \cdot a$
 C $4 \cdot a \cdot a \cdot a \cdot a$
 D $4a$

- 2 Which expression is equivalent to $\frac{10x^5}{5x^2}$? (7AF2.2)

F $2x^3$
 G $2x^7$
 H $5x^3$
 J $5x^7$

- 3 What is an expression for the area of the rectangle shown below? (7AF2.2)



A $12x^2 + 6x^4$
 B $9x^6$
 C $18x^6$
 D $12x^4 + 6x^8$

QUICK Review

STRATEGY Eliminate any answer choices that are clearly not correct.

Remember that an exponent represents how many times that the base is used as a factor.

For more help with exponents, see page 126.

READING HINT Two algebraic expressions are equivalent if one can be obtained from the other by simplifying it.

To simplify the expression, first divide 10 by 5. Then divide x^5 by x^2 .

For more help with dividing algebraic expressions, see page 544.

READING HINT To find the area of a rectangle, multiply the length by the width.

To answer this question, multiply $6x^2$ by $3x^4$ and simplify the expression.

For more help with multiplying monomials, see page 540.

QUICK Practice

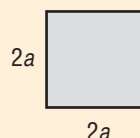
- 4 Which expression has the same value as m^{-2} ? (7AF2.1)

F m^2
 G $\frac{1}{m \cdot m}$
 H $\frac{2}{m \cdot m}$
 J $-m^2$

- 5 Which expression is equivalent to $(3y^2)(4y^3)$? (7AF2.2)

A $7y^5$
 B $7y^6$
 C $12y^5$
 D $12y^6$

- 6 Which expression is equivalent to the area of the square shown below? (7AF2.2)



F $4a^2$
 G a
 H $2a^4$
 J $4a$

- 7 $\sqrt{9y^4} =$ (7AF2.2)

A 3
 B $3y$
 C $3y^2$
 D $9y^2$

QUICK Review

READING HINT The negative exponent means multiplying by the multiplicative inverse.

To raise m to a negative exponent, first find the multiplicative inverse of m .

For more help with negative exponents, see page 127.

STRATEGY Use basic properties of multiplication to simplify an expression.

First, use the Commutative and Associative Properties of Multiplication to change the order of the variables and numbers. Then simplify.

For more help with multiplying monomials, see page 540.

READING HINT To find the area of a square, multiply the length by the width.

The length and the width of the square both have length $2a$. Multiply $2a$ by $2a$.

For more help with multiplying monomials, see page 540.

STRATEGY When taking the square root of a product, take the square root of each factor and multiply.

Using the Product Property of Square Roots, take the square root of 9 and of y^4 and then multiply.

For more help with roots of monomials, see page 553.

Practice by Standard: Algebra and Functions

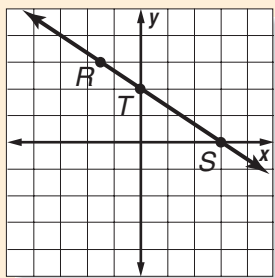
Standard Set 3.0: Students graph and interpret linear and some nonlinear functions.

DIRECTIONS

Choose the correct answer.

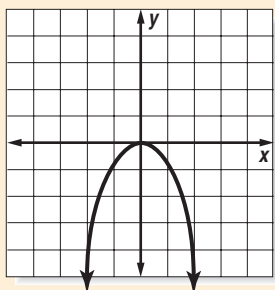
QUICK Practice

- 1 Which statement is true about the slope of line RS? (7AF3.3)



- A The slope between R and S is negative.
- B The slope between R and S is positive.
- C The slope between R and S is greater than the slope between S and T.
- D The slope is the ratio of the x - and y -intercepts.

- 2 Which equation is graphed below? (7AF3.1)



- F $y = x^2$
- G $y = -x^2$
- H $y = x^3$
- J $y = x$

QUICK Review

STRATEGY Find the slope of the line first. Then see which answer choice is correct.

When finding the slope of the line, first find two points on the line. Then, find the ratio of the difference in y -values to the difference in x -values.

For more help with slopes, see page 477.

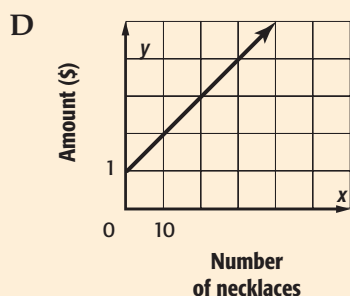
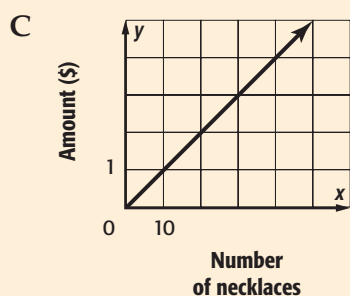
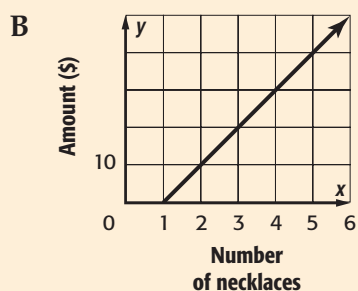
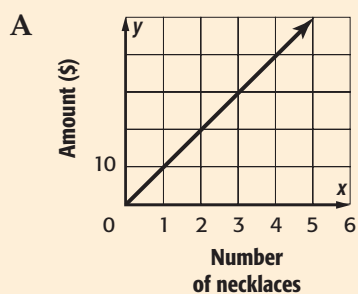
STRATEGY Think: Which answers can you eliminate because the equations do not represent quadratic functions?

Choose the equation for which coordinates of ordered pairs on the graph are solutions.

For more help with graphing, see page 522.

QUICK Practice

- 3 Kelly is going to buy some necklaces that are \$10 each. Which graph represents the number of necklaces x and the amount she will pay y ?
(7AF3.4)



QUICK Review

STRATEGY You can eliminate some choices by knowing that 0 necklaces will cost 0 dollars. As the number of necklaces increases, so will the cost.

Find a few points from the information in the question. Plot these points.

For more help with graphing linear functions, see page 472.

Practice by Standard: Algebra and Functions

Standard Set 4.0: Students solve simple linear equations and inequalities over the rational numbers.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 What value of a makes the equation below true? (7AF4.1)

$$4a - 3 = 13$$

- A 2 C 4
B 3 D 5

- 2 Miguel road his bike and jogged on Thursday. The rates and times for each activity are listed in the table below.

	Rate	Time
Biked	20 miles per hour	2 hours
Jogged	6 miles per hour	1 hour

How far did he bike and jog in all? (7AF4.2)

- F 6 miles
G 26 miles
H 46 miles
J 52 miles

- 3 What is the solution of the inequality $3x - 7 \leq 26$? (7AF4.1)

- A $x \leq 11$
B $x \geq 11$
C $x \leq \frac{19}{3}$
D $x \geq \frac{19}{3}$

QUICK Review

READING HINT The value that makes an equation true is the value that results in a true sentence.

To solve this problem, first add 3 to both sides. Then divide both sides by 4.

For more help with solving equations, see page 423.

READING HINT The question is asking what distance he traveled. Distance is found by multiplying the rate by the time.

First find the distance that he biked. Then find the distance that he jogged. Add these two distances together.

For more help with distance problems, see page 191.

READING HINT The solution is the set of all values for x that make the inequality true.

To find the solution, first add 7 to both sides, then divide both sides by 3.

For more help with solving inequalities, see page 451.

QUICK Practice

- 4 What is the value of m if $-2m + 5 = 11$?
(7AF4.1)

F -8
G -3
H 3
J 8

- 5 Jenna needs \$200 to buy a used computer. She has \$75 and can save \$25 per week. The equation $25w + 75 = 200$ shows this relationship. How many weeks must Jenna save to have the money she needs? (7AF4.1)

A 11
B 7
C 5
D 3

- 6 Margaret earns money by baby-sitting and mowing lawns. Her earnings for one week are listed in the table below.

Job	Pay
Baby-sitting	\$30 for 6 hours
Mowing Lawns	\$45 for 3 hours

She is baby-sitting on Saturday. At the rate listed above, how many hours does she need to work to earn \$50? (7AF4.2)

F 3
G 6
H 10
J 12

QUICK Review

STRATEGY Try to solve the equation using the rules of algebra. If you cannot, then substitute each answer choice into the equation to find the solution.

To solve the equation, subtract 5 from each side. Then divide each side by -2 .

For more help with solving equations, see page 423.

STRATEGY Check your answer to make sure it makes sense. For example, negative numbers would not make sense in this situation.

To solve the equation, subtract 75 from each side. Then divide your result by 25.

For more help with solving equations, see page 423.

STRATEGY Eliminate any unnecessary information. In this situation, you aren't asked anything about mowing lawns.

To solve the problem, determine Margaret's hourly rate for baby-sitting. Divide 30 by 6 to find the rate. Then divide 50 by this rate to find the number of hours.

For more help with solving rate problems, see page 191.

Practice by Standard: Measurement and Geometry

Standard Set 1.0: Students choose appropriate units of measure and use ratios to convert within and between measurement systems to solve problems.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 The table below lists the speed of several bicyclists.

Bicyclist	Speed
Kasey	22 miles in 1 hour
Brian	1 mile in 3 minutes
Tonya	1 mile in 240 seconds

Which bicyclist is the fastest? (7MG1.3)

- A Kasey
- B Brian
- C Tonya
- D not enough information

QUICK Review

READING HINT Read carefully to determine what is being asked.

Find the rates for all 3 bicyclists using common units. For example, convert all 3 rates to miles per hour.

For more help with unit rates, see page 213.

- 2 How many centimeters are in 400 millimeters? (7MG1.1)

- F 0.4
- G 40
- H 4,000
- J 40,000

STRATEGY Eliminate any answers that are unreasonable. A millimeter is smaller than a centimeter, so you would expect the number of centimeters to be less than 400.

1 centimeter is equal to 10 millimeters.
Solve the proportion $\frac{1 \text{ cm}}{10 \text{ mm}} = \frac{x \text{ cm}}{400 \text{ mm}}$.

For more help with unit conversions, see page 213.

- 3 Brandon drives his car at a rate of 75 miles per hour. How far can he drive, at this rate, in 3 hours? (7MG1.3)

- A 25 miles
- B 150 miles
- C 225 miles
- D 500 miles

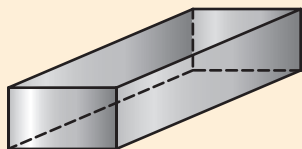
STRATEGY Try to solve a simpler problem. If he can drive 75 miles in one hour, then he can drive twice that in two hours and so on.

To find the distance he can drive in 3 hours, multiply the distance he can drive in 1 hour by 3.

For more help with the distance, rate, and time formula, see page 191.

QUICK Practice

- 4 The volume of the rectangular prism shown below is 60 cubic inches. If 1 in. = 2.54 cm, how many cubic centimeters is this? (7MG1.1)



- F 3.7
G 9.3
H 23.6
J 983.2

QUICK Review

READING HINT You are converting *cubic* inches to *cubic* centimeters. Remember, 1 inch is equal to 2.54 centimeters.

Use unit ratios.

$$60 \text{ in}^3 =$$

$$60 \cancel{\text{in.}} \times \cancel{\text{in.}} \times \cancel{\text{in.}} \times \frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \times \frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \times \frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}}$$

For more help with converting inches to centimeters, see page 221.

- 5 Which rate is equivalent to 60 miles per hour? (7MG1.1)

- A 1 mile per minute
B 1 mile per second
C 2 miles per minute
D 360 miles per minute

READING HINT 60 miles per hour is a unit rate.

To find the equivalent amount, begin with answer choice A and convert it to miles per hour. If the rates are not equivalent, keep checking.

For more help with conversions, see page 213.

- 6 Perry can ride his bike 60 miles in 3 hours. If he wants to find the rate that he rides his bike, which units will represent his rate? (7MG1.3)

- F hours per mile
G miles per hour
H miles
J hours

STRATEGY First find the answer without looking at the answer choices. If your answer does not appear in the choices recheck your work.

Rate is equal to distance divided by time.

For more help with units of rate, see page 191.

- 7 The CD Shop sells three CDs for \$43.50. Which represents the same price per CD? (7MG1.3)

- A 7 CDs for \$99.75
B 6 CDs for \$76.50
C 4 CDs for \$54.00
D 5 CDs for \$72.50

STRATEGY Think: What is the unit cost of a CD?

First find the price per CD for the CD Shop before making other comparisons.

For more help with unit rates, see page 213.

Practice by Standard: Measurement and Geometry

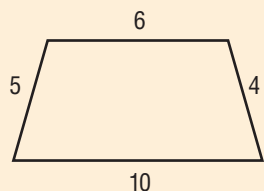
Standard Set 2.0: Students compute the perimeter, area, and volume of common geometric objects and use the results to find measures of less common objects. They know how perimeter, area, and volume are affected by changes of scale.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 What is the perimeter of the figure shown below? (7MG2.1)



- A 16 units C 25 units
B 24 units D 50 units

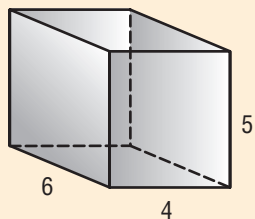
QUICK Review

READING HINT The perimeter of a figure is the distance around the figure.

To find the perimeter of the figure, add the lengths of the sides.

For more help with perimeters of irregular figures, see page 674.

- 2 What is the surface area of the solid shown below? (7MG2.3)



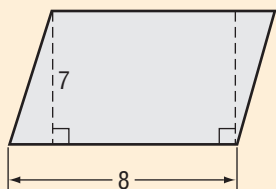
- F 52 units² H 96 units²
G 54 units² J 148 units²

READING HINT The surface area of a 3-dimensional solid is the sum of the areas of all of the faces.

Find the area of all six faces of the rectangular solid. Then add these values.

For more help with surface area of rectangular solids, see page 386.

- 3 What is the area of the parallelogram below? ($A = bh$) (7MG2.1)



- A 112 units² C 30 units²
B 56 units² D 28 units²

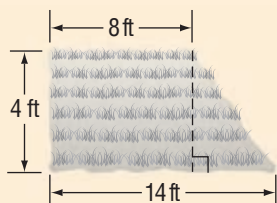
STRATEGY As you work the problem, make sure to record all of your steps so you can check your answer.

To find the area of the parallelogram, substitute 7 for the height and 8 for the base in the formula.

For more help with finding the area of a parallelogram, see page 675.

QUICK Practice

- 4 Aaron planted a garden in the shape shown below.



What is the area of the garden?

(7MG2.2)

- F 112 ft^2 H 66 ft^2
G 76 ft^2 J 44 ft^2

QUICK Review

STRATEGY Solve a simpler problem. Divide the figure into a rectangle and a triangle.

To find the area of a rectangle, use the formula $A = \ell w$. To find the area of a triangle, use the formula $A = \frac{1}{2}bh$.

For more help with finding the area of irregular figures, see page 674.

- 5 Laura wants to cover her outdoor pool that has a diameter of 15 feet. What is the area of the pool?

($A = \pi r^2$, use 3.14 for π) (7MG2.1)

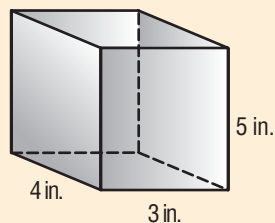
- A 23.6 ft^2
B 47.1 ft^2
C 176.6 ft^2
D 706.5 ft^2

READING HINT You are looking for the area of the pool.

To find the area, substitute the radius into the formula for the area of a circle. Square that value and then multiply the result by π . Remember that the radius is one half of the diameter.

For more help with finding the area of a circle, see page 354.

- 6 What would happen to the surface area of the rectangular solid shown below if all 3 dimensions are multiplied by 2? (7MG2.3)



- F It would be multiplied by 2.
G It would be multiplied by 4.
H It would be multiplied by 8.
J It would remain the same.

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

To find the surface area, find the area of all 6 faces. Then, double the dimensions and find the new surface area. Finally, compare these values. It is easier to solve by noting that since the length of each side is increased by a factor of 2, the surface area is increased by a factor of $(2)^2$, or 4.

For more help with surface area, see page 400.

Practice by Standard: Measurement and Geometry

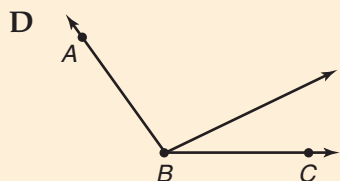
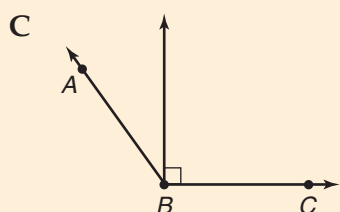
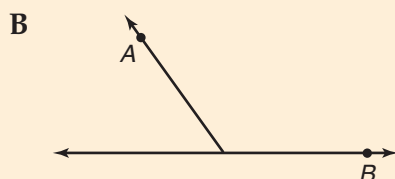
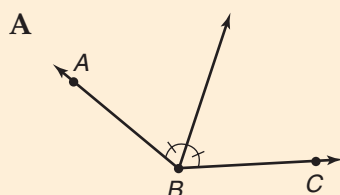
Standard Set 3.0: Students know the Pythagorean Theorem and deepen their understanding of plane and solid geometric shapes by constructing figures that meet given conditions and by identifying attributes of figures.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 Which answer choice shows the angle bisector of angle ABC ? (7MG3.1)



- 2 Which of the following is *not* a possible way that three planes can intersect? (7MG3.6)
- F They can intersect in a line.
 - G They can intersect at a plane.
 - H They can intersect at a point.
 - J They can have no intersection.

QUICK Review

READING HINT An angle bisector is a line that divides an angle into two congruent angles.

To find the angle bisector, find the answer choice in which two angles have the same measure.

For more help with angle bisectors, see page 312.

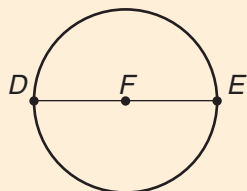
STRATEGY Try drawing a picture of the situation. Sometimes diagrams can be helpful.

From your drawing, try to eliminate any choices that you know *are* possible ways that three planes can intersect.

For more help with intersecting planes, see page 365.

QUICK Practice

- 3 In the circle below, F is the center. Line segment DF shown below is called a— (7MG3.1)



- A radius. C chord.
B diameter. D central angle.

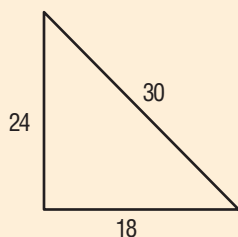
QUICK Review

READING HINT You are examining a line that goes through the center of the circle.

DF is only half of the chord that goes through the center of the circle.

For more help with naming parts of circles, see page 352.

- 4 Which statement below can be used to prove that the triangle is a right triangle? (7MG3.3)



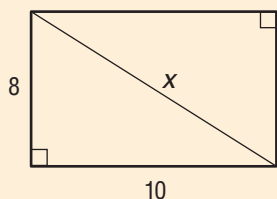
- F $24 + 18 = 30$
G $24 + 18 > 30$
H $24^2 + 18^2 = 30^2$
J $24^2 + 18^2 > 30^2$

READING HINT In a right triangle, the sum of the square of the legs is equal to the square of the hypotenuse.

You can eliminate two of the answer choices since they don't involve squares. Check to see if $24^2 + 18^2$ is equal to 30^2 .

For more help with the Pythagorean Theorem, see page 162.

- 5 What is the value of x in the diagram below? (7MG3.3)



- A 18 C 6.0
B 12.8 D 2.0

STRATEGY You can use the Pythagorean Theorem to solve this problem because x is part of a right triangle.

Solve the equation $8^2 + 10^2 = x^2$.

For more help with the Pythagorean Theorem, see page 162.

Practice by Standard: Statistics, Data Analysis, and Probability

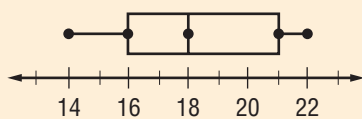
Standard Set 1.0: Students collect, organize, and represent data sets that have one or more variables and identify relationships among variables within a data set by hand and through the use of an electronic spreadsheet software program.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 The box-and-whisker plot represents the age of customers at a music store.



Ages at the Music Store

What was the median age?
(7SDAP1.3)

- A 16 C 21
B 18 D 22

QUICK Review

READING HINT The median is the middle number of the data set.

On a box-and-whisker plot, the line inside the box represents the median.

For more help with medians, see page 585.

- 2 The table below shows the number of hours that 8 seventh graders spent watching television last week.

2	2	5	3
3	4	6	4

What is the maximum of the data set? (7SDAP1.3)

- F 2 H 5
G 4 J 6

READING HINT The maximum is the greatest number of a data set.

To find the maximum of the data set, find the largest value.

For more help with maximums of data sets, see page 593.

- 3 What is the minimum of the data set below? (7SDAP1.3)

35, 32, 14, 18, 32, 38, 40, 52

- A 10 C 18
B 14 D 32

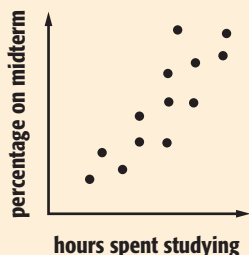
READING HINT The minimum of a data set is the smallest number of a data set.

To find the minimum, find the smallest number.

For more help with minimums of data sets, see page 593.

QUICK Practice

- 4 The scatter plot shows the number of hours that students spent studying for a midterm exam and the grade they received on the exam.



Which statement *best* describes the relationship between the number of hours spent studying and the grade received on the midterm?

(7SDAP1.2)

- F As hours increase, midterm grade increases.
- G As hours decrease, midterm grade increases.
- H As hours increase, midterm grade increases, then decreases.
- J As hours increase, midterm grade decreases, then increases.

QUICK Review

READING HINT You are looking for the statement that *best* describes the data.

Looking at the scatter plot, you can see there is a positive correlation.

For more help with interpreting scatter plots, see page 505.

- 5 Cecilia surveyed 10 students. She asked them how long, in minutes, it takes them to get to school in the morning. She recorded the data in the table below.

25	5	35	45	16
14	10	22	20	12

What is the lower quartile of the data? (7SDAP1.3)

- A 5
- B 10
- C 12
- D 16

READING HINT The lower quartile is the median of the lower half of the data set.

To find the lower quartile, first order the data. Then find the median of the lower half.

For more help with finding the lower quartile, see page 593.

Practice by Standard: Mathematical Reasoning

Standard Set 1.0: Students make decisions about how to approach problems.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 Piper has \$200 in a savings account. She wants to save \$500. If she can save \$50 per week, how many weeks will it take her to save \$500? (7MR1.1, 7AF4.1)
- A 10
 - B 8
 - C 6
 - D 4

QUICK Review

STRATEGY See if you can eliminate any answers because they are not reasonable.

You can write and solve an equation to solve this. Use the equation $200 + 50w = 500$.

For more help with writing equations, see page 57.

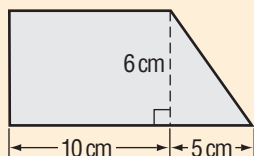
- 2 If the pattern continues, what will be the next number? (7MR1.1, 7NS1.2)
1, 3, 9, 27 ...
- F 81
 - G 54
 - H 9
 - J 3

READING HINT You are only looking for the next number.

Look at the pattern, the next number is found by multiplying the previous number by 3.

For more help with patterns, see page 124.

- 3 What is the area of the figure below? (7MR1.3, 7MG2.2)



- A 90 square centimeters
- B 85 square centimeters
- C 75 square centimeters
- D 60 square centimeters

STRATEGY Try to break the problem into simpler parts.

To find the total area, find the area of the rectangle and add that to the area of the triangle.

For more help with finding the area of polygons, see page 360.

Practice by Standard: Mathematical Reasoning

Standard Set 2.0: Students use strategies, skills, and concepts in finding solutions.

DIRECTIONS

Choose the correct answer.

QUICK Practice

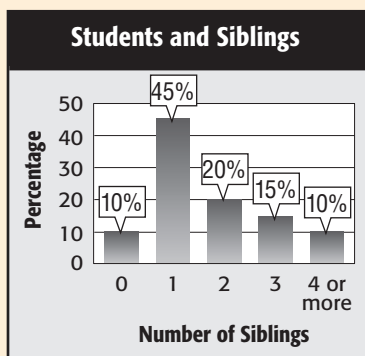
- 1 Karen is filling up a child's sandbox with sand. Which estimate is a reasonable amount of sand that she will need? (7MR2.1, 7MG1.3)

A 3 ft³ C 350 ft³
B 35 ft³ D 3,500 ft³

- 2 Ely is baking cookies for a school carnival. It takes 2 cups of flour for one batch. If he wants to make 10.5 batches, how much flour will he need? (7MR2.2, 7NS1.2)

F 5.25 cups H 21.0 cups
G 12.0 cups J 28.0 cups

- 3 The graph below shows the percentage of students that have each number of siblings.



If 50 students were surveyed, how many would you expect to have 4 or more siblings? (7MR2.3, 7SDAP1.1)

A 1 C 10
B 5 D 25

QUICK Review

READING HINT You are looking for an estimate. Try to draw a picture of a sample sandbox.

The formula for the volume of a rectangular prism is $V = \ell wh$.

For more help with estimating volume, see page 373.

STRATEGY Try to solve this by solving a simpler problem. It takes 2 cups for one batch; thus, it will take 4 cups for 2 batches and so on.

You can find the amount of flour by setting up and solving this proportion

$$\frac{2 \text{ cups}}{1 \text{ batch}} = \frac{x \text{ cups}}{10.5 \text{ batches}}$$

For more help with proportions, see page 194.

STRATEGY Make sure to compare the labels on the graph with what is being asked.

The graph shows that 10% of the students have 4 or more siblings. Find 10% of 50.

For more help with finding percentages of numbers, see page 263.

Practice by Standard: Mathematical Reasoning

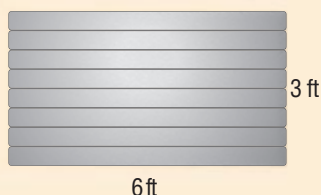
Standard Set 3.0: Students determine a solution is complete and move beyond a particular problem by generalizing to other situations.

DIRECTIONS

Choose the correct answer.

QUICK Practice

- 1 Alex wants to cover 6 table tops like the one shown below with fabric. Which amount of fabric is a reasonable estimate for how much he will need? (7MR3.1, 7MG2.1)



- A 60 ft³ C 200 ft³
B 120 ft³ D 500 ft³

QUICK Review

READING HINT To cover the tables you will need to find the area of the tops of the tables.

First find the area of the top of one table. Remember that $A = \ell w$. Then multiply your estimate by 6. Since you are estimating, round the area to 20.

For more help with areas of rectangles, see page 674.

- 2 A rafting company charges \$50 to rent a raft plus \$15 per hour. How much will it cost to rent the raft for 8 hours? (7MR3.2, 7AF4.2)

- F \$65 H \$170
G \$120 J \$415

STRATEGY You can generalize the situation by writing an equation.

First, find the cost for the 8 hours by multiplying 8 times 15. Add this amount to 50, the amount to rent the raft.

For more help with writing equations, see page 57.

- 3 If the pattern continues with the numbers below, what will be the next number in the pattern? (7MR3.3, 7NS1.2)

1, 6, 16, 31, 51, ...

- A 61
B 71
C 76
D 80

STRATEGY When looking at a pattern, try to see if there is either a pattern in the differences or a common factor.

In this data set, the difference between the first 2 numbers is 5 (5×1). The difference between the second 2 numbers is 10 (5×2). The difference between the next 2 numbers is 15 (5×3). The next difference is 20 (5×4). You would expect the following difference to be 25, and the next number in the pattern to be $51 + 25 = 76$.

For more help with finding patterns, see page 124.